

Chapter 2. Hilbert Spaces.

→ the fundamental mathematical framework for quantum mechanics.

• quantum state spaces: complex vector spaces.

eg: $H_1 = \{ \alpha |0\rangle + \beta |1\rangle, \alpha, \beta \in \mathbb{C} \}$. $|0\rangle, |1\rangle$ — kets.

Basis: $B_1 = (|0\rangle, |1\rangle)$.

H_1 is the set of all formal linear combinations of the elements of B_1 with complex coefficients.

More general: Basis: $B_n = (|b\rangle)_{b \in \{0,1\}^n}$.

① $H_n = \sum_{b \in \{0,1\}^n} \mathbb{C} |b\rangle$. $\dim(H_n) = 2^n$.

Since $\dim(H_n) = 2^n$, we can rewrite $B_n = (|0\rangle_n, |1\rangle_n, \dots, |2^n-1\rangle_n)$
 $\triangleq (|b\rangle_n)_{b \in \mathbb{Z}_{2^n}}$.

and ② $H_n = \left\{ \sum_{b=0}^{2^n-1} \alpha_b |b\rangle_n : \alpha_b \in \mathbb{C} \text{ with } b \in \mathbb{Z}_{2^n} \right\}$.

③ vector representations.

For k -dimension complex vector space H , basis $B \triangleq (|b_0\rangle, \dots, |b_{k-1}\rangle)$.

then each $|\psi\rangle \in H$, we can write $|\psi\rangle = \alpha_0 |b_0\rangle + \dots + \alpha_{k-1} |b_{k-1}\rangle$.

isomorphism $H \longleftrightarrow \mathbb{C}^k$.

$$|\psi\rangle \longleftrightarrow (\alpha_0, \dots, \alpha_{k-1}) \triangleq |\psi\rangle_B.$$

$$\Rightarrow |\psi\rangle = (|b_0\rangle, \dots, |b_{k-1}\rangle) \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{k-1} \end{pmatrix}.$$

eg: $|\psi\rangle = |0\rangle + i|1\rangle$, $B = (|0\rangle, |1\rangle)$.

then $|\psi\rangle_B = (1, i)$.

2°. Inner products.

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bra.

$$\langle \cdot | \cdot \rangle : H \times H \rightarrow \mathbb{C}.$$

$$(|\psi\rangle, |\varphi\rangle) \mapsto \langle \psi | \varphi \rangle.$$

$$\langle \varphi | \in H^*.$$

$$\langle \varphi | : H \rightarrow \mathbb{C}.$$

$$|\varphi\rangle \mapsto \langle \varphi | \varphi \rangle.$$

$$|\psi\rangle = \sum_{i=0}^{m-1} \alpha_i |\varphi_i\rangle, \quad |\varphi\rangle = \sum_{i=0}^{m-1} \beta_i |\varphi_i\rangle.$$

Isomorphism.

then.
$$\langle \varphi | \varphi \rangle = \sum_{i=0}^{m-1} \sum_{j=0}^{m-1} \overline{\alpha_i} \beta_j \langle \varphi_i | \varphi_j \rangle.$$

$\swarrow$
 complex conjugate of α_i

★ Inner product on H with respect to the basis $B = (|b_0\rangle, \dots, |b_{k-1}\rangle)$.

$$\langle b_i | b_j \rangle_B = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}.$$

then. for. $|\psi\rangle = (|b_0\rangle, \dots, |b_{k-1}\rangle) \cdot \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{k-1} \end{pmatrix} = |\psi\rangle_B$ $|\varphi\rangle = (|b_0\rangle, \dots, |b_{k-1}\rangle) \cdot \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{k-1} \end{pmatrix} = |\varphi\rangle_B$

we have
$$\langle \varphi | \psi \rangle = (\overline{\alpha_0} \dots \overline{\alpha_{k-1}}) \cdot \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{k-1} \end{pmatrix} = |\varphi\rangle_B^* \cdot |\psi\rangle_B$$

Bra-ket notation (Dirac notation).

introduced by Paul Dirac
1930s.

$\swarrow$
 dual vectors $\searrow$
 state vector.

$(H, \langle \cdot | \cdot \rangle)$ — Hilbert space.

Norm.
$$\| |\psi\rangle \| \triangleq \| \psi \| = \sqrt{\langle \psi | \psi \rangle}.$$



Orthogonality:

★ $|\psi\rangle$ and $|\varphi\rangle$ are orthogonal to each other $\Leftrightarrow \langle \varphi | \psi \rangle = 0$.

★ orthonormal sequence B . $\forall |\psi\rangle, |\varphi\rangle \in B. \Rightarrow \langle \varphi | \psi \rangle = 0, \| \psi \| = \| \varphi \| = 1$.

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$B = \{|b_0\rangle, \dots, |b_{k-1}\rangle\}$ is an orthonormal basis of $(H, \langle \cdot | \cdot \rangle_B)$.

$$|\psi\rangle = \sum_{i=0}^{k-1} \langle b_i | \psi \rangle |b_i\rangle.$$

3°. Linear maps: $H \rightarrow H'$
 $\mathbb{C}^k \quad \mathbb{C}^l$
 basis $B \quad C$

$$f \in \text{Hom}(H, H') \longrightarrow \text{Mat}_{B,C}(f) \in \mathbb{C}^{k \times l}.$$

eg: Pauli X operator on H_1 . $B = (|0\rangle, |1\rangle)$

$$X: H_1 \rightarrow H_1. \quad \alpha|0\rangle + \beta|1\rangle \longmapsto \beta|0\rangle + \alpha|1\rangle.$$

$$\text{Set } \text{Mat}_B(X) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ then } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}.$$

$$\Rightarrow \begin{cases} a\alpha + b\beta = \beta \\ \alpha + d\beta = \alpha \end{cases} \Rightarrow \begin{cases} a=0 \\ b=1 \\ c=1 \\ d=0 \end{cases} \text{ ie. } \text{Mat}_B(X) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$\text{basis } C = (|x_+\rangle, |x_-\rangle) = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

$$\Rightarrow \text{Mat}_C(X) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{different basis} \Rightarrow \text{different Matrix representation}$$

$$\left\{ \begin{array}{l} \text{Pauli Z operator: } \text{Mat}_B(Z) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \text{Pauli Y operator: } \text{Mat}_B(Y) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ \text{Hadamard operator: } H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{array} \right.$$

Quantum gates.

$$\text{Prop: } \langle \psi | T | \varphi \rangle = \langle \varphi | T^\dagger | \psi \rangle. \quad \text{Mat}_{B,C}(T) = (\langle c_i | T | b_j \rangle)_{i \in \mathbb{Z}_l, j \in \mathbb{Z}_k} \in \mathbb{C}^{l \times k}.$$

4°. The Hilbert-Schmidt inner product. $(\mathbb{C}^{l \times k})$

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$$\langle \cdot | \cdot \rangle : \mathbb{C}^{(l \times k)} \times \mathbb{C}^{(l \times k)} \rightarrow \mathbb{C}$$

$$(A, B) \mapsto \langle A | B \rangle = \text{tr}(A^* B)$$

$$A^* = \overline{A}^T \in \mathbb{C}^{k \times l}$$

$$A = (a_0, \dots, a_{k-1}) \quad B = (b_0, \dots, b_{k-1})$$

$$\Rightarrow A^* = \begin{pmatrix} a_0^* \\ \vdots \\ a_{k-1}^* \end{pmatrix} \Rightarrow A^* B = \begin{pmatrix} a_0^* b_0 & a_0^* b_1 & \dots & a_0^* b_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k-1}^* b_0 & a_{k-1}^* b_1 & \dots & a_{k-1}^* b_{k-1} \end{pmatrix}$$

$$\Rightarrow \text{tr}(A^* B) = \sum_{i=0}^{k-1} a_i^* b_i$$

5°. Endomorphism.

$$\mathbb{C}^{k \times k} : H \rightarrow H$$

$$A \in \mathbb{C}^{k \times k} \Leftrightarrow A \in \text{End}(H)$$

eg: Pauli operators X, Y, Z on H , with respect to $\underbrace{(|0\rangle, |1\rangle)}_{\text{basis}}$ are Pauli matrices.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Hadamard operator on H , with respect to $(|0\rangle, |1\rangle)$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

involutions

P73. check properties of matrix $A \in \mathbb{C}^{(k \times k)}$

$$A \in \mathbb{C}^{(k \times k)}$$

or $A \in \text{End}(H)$ is called an involution if $A^2 = I_k$ or $A^2 = I_H$.

6. Hermitian matrices or operators. $\Leftrightarrow A = A^*$

Unitary $\swarrow$ used to model the evolution of quantum systems over time.
matrices or operators $\Leftrightarrow U^* U = U U^* = I_k$.

eg: Pauli X, Y, Z . Hadamard operator H .

Rows or columns form an orthonormal basis of $\mathbb{C}^k$.

7°. Outer products.

$B = (|b_0\rangle, \dots, |b_{k-1}\rangle)$ be an orthonormal basis of H .

$$|\psi\rangle\langle\psi| : H \rightarrow H$$

$$\in \text{End}(H) \quad |z\rangle \mapsto |\psi\rangle \cdot \underbrace{\langle\psi|z\rangle}_{\in \mathbb{C}}$$

density operators

P_{123}

$$\text{Mat}_B(|\psi\rangle\langle\psi|) = (\alpha_i \bar{\beta}_j)$$

$$= \begin{pmatrix} \alpha_1 \bar{\beta}_1 & \dots & \alpha_1 \bar{\beta}_k \\ \vdots & & \vdots \\ \alpha_k \bar{\beta}_1 & & \alpha_k \bar{\beta}_k \end{pmatrix}$$

eg: H , single-qubit state space with Basis $(|0\rangle, |1\rangle)$.

Example of outer product of kets in H are $|0\rangle\langle 0|$, $|0\rangle\langle 1|$, $|1\rangle\langle 0|$, $|1\rangle\langle 1|$.

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \in H$, then

$$|0\rangle\langle 0| (\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle + \beta\langle 0|1\rangle = \alpha|0\rangle$$

$$|0\rangle\langle 1| (\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle$$

$$|1\rangle\langle 0| (\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle$$

$$|1\rangle\langle 1| (\alpha|0\rangle + \beta|1\rangle) = \beta|1\rangle$$

8°. Projections. — measurement.

Def: $P \in \text{End}(H)$ is called a projection if $P^2 = P$.

A projection $P \in \text{End}(H)$ is called orthogonal if

$P|\psi\rangle$ and $|\psi\rangle - P|\psi\rangle$ are orthogonal to each other.

Prop: Projection P : orthogonal $\Leftrightarrow$ hermitian.

eg: $P_i : H \rightarrow H(i)$. $|\psi\rangle \mapsto |\psi(i)\rangle$. $(|\psi\rangle = \sum_{i=0}^{L-1} |\psi(i)\rangle)$

9°. Important decompositions.

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①. Schur decomposition.

$A \in \mathbb{C}^{k,k}$. eigenvalues $\lambda_0 \dots, \lambda_{l-1}$
algebraic multiplicities $m_0 \dots m_{l-1}$

then. $\exists U \in \mathbb{C}^{k,k}$ — unitary.

$T \in \mathbb{C}^{k,k}$ — upper triangular matrix

$$T = \begin{pmatrix} \lambda_0 & & & * \\ & \lambda_0 & & \\ & 0 & \lambda_1 & \\ & & \ddots & \ddots \\ & & & \lambda_{l-1} \end{pmatrix}$$

s.t. $A = UTU^*$.

②. The spectral theorem. $\rightarrow$ postulates of quantum mechanics.

Def: $A \in \text{End}(H)$ normal $\Leftrightarrow A^*A = AA^*$.

Thm: $A \in \mathbb{C}^{k,k}$ normal.

$\lambda_0 \dots \lambda_{l-1}$ eigenvalues

$m_0 \dots m_{l-1}$ algebraic multiplicities.

$\Rightarrow \exists U \in \mathbb{C}^{k,k}$ unitary. s.t.

$$U^*AU = \begin{pmatrix} \lambda_0 & & & \\ & \lambda_0 & & \\ & & \lambda_1 & \\ & & & \ddots \\ & & & & \lambda_{l-1} \end{pmatrix}$$

$$U = \left(\underbrace{\vec{u}_{m_0} \dots \vec{u}_{m_1-1}}_{\text{orthonormal basis of eigenspace with } \lambda_0} \quad \underbrace{\vec{u}_{m_1} \dots \vec{u}_{m_2-1}}_{\text{orthonormal basis of eigenspace with } \lambda_1} \quad \dots \quad \underbrace{\vec{u}_{m_{l-1}} \dots \vec{u}_{m_l-1}}_{\text{orthonormal basis of eigenspace with } \lambda_{l-1}} \right)$$

orthonormal basis

of eigenspace with λ_i .

* Then (Spectral theorem).

* Singular value decomposition.

$$A \in \mathbb{C}^{(k,l)} \quad \text{rank}(A) = r.$$

then, $\exists U \in \mathbb{C}^{k,k}, V \in \mathbb{C}^{l,l}$ unitary matrices.

s.t.

$$A = U \left(\begin{array}{c|c} \lambda_0 \dots \lambda_{r-1} & 0 \\ \hline 0 & 0 \end{array} \right) V^*.$$

where $\lambda_0 \dots \lambda_{r-1}$ are positive real numbers.
unique.

Tensor Products. (key role in quantum mechanics) 8

- ① mathematical description of composite quantum systems.
- ② distinction between separable and entangled states qubits.
- ③ construction of quantum gates and quantum circuits.
- ④ formulation of density matrices for mixed states.

Example 2.5.1.

$$H_1 = \sum_{b \in \{0,1\}} \mathbb{C} |b\rangle. \quad B_1 = (|0\rangle, |1\rangle).$$

$$\Rightarrow H = H_1 \otimes H_1, \text{ with basis } B = (|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle) \\ = (|00\rangle, |01\rangle, |10\rangle, |11\rangle) \\ = (|0\rangle|0\rangle, |0\rangle|1\rangle, |1\rangle|0\rangle, |1\rangle|1\rangle)$$

$$\dim(H) = 4.$$

More general.

$$H = H(0) \otimes H(1) \otimes \dots \otimes H(m-1).$$

$$\begin{array}{ccc} \dim \downarrow & \downarrow & \downarrow \\ k_0 & k_1 & k_m \\ \text{Basis } B_0 & B_1 & B_m. \end{array}$$

$$\Rightarrow \dim(H) = K = \prod_{j=0}^{m-1} k_j. \quad B = B_0 \otimes \dots \otimes B_{m-1}.$$

$$\mathbb{Z}_{\vec{K}} = \prod_{j=0}^{m-1} \mathbb{Z}_{k_j} \text{ with } \vec{K} = (k_0, \dots, k_{m-1}).$$

$$\text{For } \vec{i} \in \mathbb{Z}_{\vec{K}}, \quad \vec{i} = (i_0, \dots, i_{m-1}), \text{ with } i_j \in \mathbb{Z}_{k_j}.$$

$$|b_{\vec{i}}\rangle = \bigotimes_{j=0}^{m-1} |b_{i_j, j}\rangle = |b_{i_0, 0}\rangle \dots |b_{i_{m-1}, m-1}\rangle. \quad |b_{i_j, j}\rangle \in H_j.$$

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Calculate with Basis.

Prop 2.5.3. $0 \leq j \leq m$, $|\varphi_j\rangle \in H(j)$. with $|\varphi_j\rangle = \sum_{i=0}^{k_j-1} \alpha_{i,j} \underline{b_{i,j}}$

Then. $\bigotimes_{j=0}^{m-1} |\varphi_j\rangle = \sum_{\vec{i} \in \mathbb{Z}_K} \alpha_{\vec{i}} |b_{\vec{i}}\rangle$ $\vec{i} = (i_0, \dots, i_{m-1}) \in \mathbb{Z}_K$. basis of $H(j)$.

and. $\alpha_{\vec{i}} = \prod_{j=0}^{m-1} \alpha_{i_j, j}$ $\approx |b_{\vec{i}}\rangle$

Proof: $|b_{i_0,0}\rangle \dots |b_{i_{j-1},j-1}\rangle \dots |b_{i_{m-1},m-1}\rangle \alpha_{i_0,0} \dots \alpha_{i_{j-1},j-1} \dots \alpha_{i_{m-1},m-1} \triangleq \alpha_{\vec{i}}$.

so. $\bigotimes_{j=0}^{m-1} |\varphi_j\rangle = \sum_{\vec{i} \in \mathbb{Z}_K} \alpha_{\vec{i}} |b_{\vec{i}}\rangle$.

2.5.2. Inner product.

$H = H(0) \otimes \dots \otimes H(m-1)$. $B = B_0 \otimes \dots \otimes B_{m-1}$.

Prop 2.5.5. $\langle \bigotimes_{j=0}^{m-1} |\varphi_j\rangle | \bigotimes_{j=0}^{m-1} |\varphi_j\rangle \rangle = \prod_{j=0}^{m-1} \langle \varphi_j | \varphi_j \rangle$.

Proof: ① let. $|\varphi_j\rangle = \sum_{i=0}^{k_j-1} \alpha_{i,j} \underbrace{|b_{i,j}\rangle}_{\text{basis}}$. $|\varphi_j\rangle = \sum_{i=0}^{k_j-1} \beta_{i,j} |b_{i,j}\rangle$.

$\Rightarrow |\varphi\rangle = \sum_{\vec{i} \in \mathbb{Z}_K} \alpha_{\vec{i}} |b_{\vec{i}}\rangle$. $|\varphi\rangle = \sum_{\vec{i} \in \mathbb{Z}_K} \beta_{\vec{i}} |b_{\vec{i}}\rangle$.

$\Rightarrow \langle \varphi | \varphi \rangle = \langle \sum_{\vec{i} \in \mathbb{Z}_K} \alpha_{\vec{i}} |b_{\vec{i}}\rangle | \sum_{\vec{i} \in \mathbb{Z}_K} \beta_{\vec{i}} |b_{\vec{i}}\rangle \rangle$
 $= \sum_{\vec{i} \in \mathbb{Z}_K} \overline{\alpha_{\vec{i}}} \cdot \beta_{\vec{i}}$.

②. $\prod_{j=0}^{m-1} \langle \varphi_j | \varphi_j \rangle = \prod_{j=0}^{m-1} \langle \sum_{i=0}^{k_j-1} \alpha_{i,j} |b_{i,j}\rangle | \sum_{i=0}^{k_j-1} \beta_{i,j} |b_{i,j}\rangle \rangle$
 $= \prod_{j=0}^{m-1} \sum_{i=0}^{k_j-1} \overline{\alpha_{i,j}} \beta_{i,j} = \sum_{i=0}^{k_j-1} \overline{\alpha_{\vec{i}}} \beta_{\vec{i}}$.

$$H = H_{n_0} \otimes \dots \otimes H_{n_{m-1}} \quad \text{with} \quad n = \sum_{j=0}^{m-1} n_j.$$

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$$H \rightarrow H_n \quad |\vec{b}_0\rangle \dots |\vec{b}_{m-1}\rangle \mapsto |\vec{b}_0 \vec{b}_1 \dots \vec{b}_{m-1}\rangle \quad \underline{\text{isometry}}.$$

eg: $|0\rangle|0\rangle = |00\rangle.$

2.5.4. Homomorphisms. (quantum gate. circuits).

~~H'_j is the dual space of H_j~~

Matrix representation.

$$\text{Hom} \left(\underset{B}{H}, \underset{C}{H'} \right) \rightarrow C^{(l,k)}. \quad f \mapsto \text{Mat}_{B,C}^k(f).$$

$$f_0 \in H_0.$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\begin{matrix} f_0 & f_1 \\ A \in C^{k_0, k'_0} & B \in C^{k_1, k'_1} \end{matrix}$$

$$\begin{matrix} f \in \text{Hom} \left(\underset{B}{H}, \underset{B'}{H'} \right) \\ \parallel & \parallel \\ H_0 \otimes H & H'_0 \otimes H'_1 \\ & A \end{matrix}$$

$$f: \otimes |\varphi_0\rangle \otimes |\varphi_1\rangle \mapsto \underbrace{f_0(|\varphi_0\rangle)} \otimes f_1(|\varphi_1\rangle).$$

$$A |\varphi_0\rangle \otimes B |\varphi_1\rangle.$$

$$(A) \otimes (B) \mapsto (A \otimes B)$$

$$\psi \otimes \gamma$$

$$f = (f_0, f_1)$$

$$f_0 \in \text{Hom}(H(0), H(0'))$$

$$A_0 \in C^{k_0, k'_0}$$

$$A_1 \in C^{k_1, k'_1}$$

$$\psi \in H(0), \quad A_0 \psi \in C^{k'_0}$$

$$\gamma \in H(1), \quad A_1 \gamma \in C^{k'_1}$$

$$H(0) \otimes H(1) \longrightarrow H'(0) \otimes H'(1)$$

$$|\psi\rangle \otimes |\gamma\rangle \longrightarrow |A_0 \psi\rangle \otimes |A_1 \gamma\rangle$$

$$B(|\psi\rangle \otimes |\gamma\rangle) = \dots \otimes \dots$$

$$A_0 \otimes A_1$$

$$A_0 \otimes A_1 = \begin{pmatrix} a_{11} A_1 & a_{12} A_1 \\ a_{21} A_1 & a_{22} A_1 \end{pmatrix} \begin{pmatrix} \psi \\ \gamma \end{pmatrix} = \begin{pmatrix} a_{11} b_1 \psi + a_{12} b_2 \gamma \\ a_{21} b_1 \psi + a_{22} b_2 \gamma \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} b_1 \psi + a_{12} b_2 \gamma \\ a_{21} b_1 \psi + a_{22} b_2 \gamma \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} b_1 + a_{12} b_2 \\ a_{21} b_1 + a_{22} b_2 \end{pmatrix} \otimes A_1 \gamma$$

$$= A_0 \psi \otimes A_1 \gamma$$

Hadamard operator. $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$(H \otimes H)(|0\rangle \otimes |0\rangle) = H|0\rangle \otimes H|0\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$= \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle$$

calculate eigenvalue.

$$A_0 |\psi_0\rangle = \lambda_0 |\psi_0\rangle$$

$$A_{m-1} |\psi_{m-1}\rangle = \lambda_{m-1} |\psi_{m-1}\rangle$$

$$A \left(\bigotimes_{j=0}^{m-1} |\psi_j\rangle \right) = A_0 |\psi_0\rangle \otimes \dots \otimes A_{m-1} |\psi_{m-1}\rangle$$

$$= \lambda_0 \dots \lambda_{m-1} |\psi_0\rangle \otimes \dots \otimes |\psi_{m-1}\rangle$$

Example 2.5.17.

$H = H_1 \otimes H_1$. A_0, A_1 — Pauli X operator

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$|x_+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \quad |x_-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$X(|x_+\rangle) = |x_+\rangle \quad X(|x_-\rangle) = -|x_-\rangle$$

projection. $|x_+\rangle \langle x_+| \quad |x_-\rangle \langle x_-|$

$$A = A_0 \otimes A_1$$

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the eigenvalues of A are 1 and -1 .

$$A |x_+ x_+\rangle = A_0 |x_+\rangle \otimes A_1 |x_+\rangle = |x_+\rangle \otimes |x_+\rangle = |x_+ x_+\rangle$$

$$A |x_- x_-\rangle = A_0 |x_-\rangle \otimes A_1 |x_-\rangle = (-1)|x_-\rangle \otimes (-1)|x_-\rangle = |x_- x_-\rangle$$

$$A |x_+ x_-\rangle = -|x_+ x_-\rangle \quad A |x_- x_+\rangle = -|x_- x_+\rangle$$

eigenspaces

$$E_1 \quad B_1 = (|x_+ x_+\rangle, |x_- x_-\rangle)$$

$$E_{-1} \quad B_{-1} = (|x_+ x_-\rangle, |x_- x_+\rangle)$$

Projection

$$P_1 = |x_+ x_+\rangle \langle x_+ x_+| + |x_- x_-\rangle \langle x_- x_-|$$

$$P_2 = |x_+ x_-\rangle \langle x_- x_+| + |x_- x_+\rangle \langle x_+ x_-|$$

Thm. 2.5.18. (Schmidt decomposition theorem)

$$|\psi\rangle \in H(0) \otimes H(1) \quad B_0 = (|u_0\rangle, \dots, |u_{m-1}\rangle)$$

$$B_1 = (|v_0\rangle, \dots, |v_{m-1}\rangle)$$

then $|\psi\rangle$ can be written (uniquely).

$$|\psi\rangle = \sum_{i=0}^{m-1} r_i |u_i\rangle \otimes |v_i\rangle \quad \text{with } r_i \text{ real numbers positive.}$$

there are

Example: $|\psi\rangle = |0\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} \in H_1 \otimes H_1$ separable.

BAC

$$|\psi\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \in H(0) \otimes H(1)$$

$$|\psi\rangle = BAC.$$

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$$A = U \begin{pmatrix} D \\ 0 \end{pmatrix} V^* \quad (\text{singular-value decomposition})$$

$$\downarrow$$

$$\in \mathbb{C}^{k,k}$$

Unitary

$$\downarrow$$

$$\in \mathbb{C}^{l,l}$$

Unitary

D: positive semidefinite diagonal matrix.

$$D = (\gamma_0, \dots, \gamma_{l-1}) \quad \gamma_i \geq 0 \in \mathbb{R}.$$

$$U = (U_1, U_2)$$

$$\in \mathbb{C}^{k,l} \quad \in \mathbb{C}^{k,k-l}$$

$$\Rightarrow A = (U_1, U_2) \begin{pmatrix} D \\ 0 \end{pmatrix} V^*$$

$$= U_1 D V^*$$

$$\Rightarrow |\psi\rangle = \underbrace{B U_1}_{\text{check}} D V^* C$$

$$\leftarrow B U_1 = (|b_0\rangle, \dots, |b_k\rangle) \begin{pmatrix} | \\ | \end{pmatrix}_{k \times l} = (|u_0\rangle, \dots, |u_{l-1}\rangle)$$

$$V^* C = \begin{pmatrix} \text{---} \end{pmatrix}_{l \times l} \cdot (|c_0\rangle, \dots, |c_l\rangle)^T = (|v_0\rangle, \dots, |v_{l-1}\rangle)^T$$

$$\Rightarrow |\psi\rangle = \sum_{i=0}^{l-1} \gamma_i |u_i\rangle \otimes |v_i\rangle$$